Write a program for performing industrial data analysis using relevant tools and techniques

Turbine Fault IIoT Data

Dataset Description

This dataset represents Industrial IoT (IIoT) sensor logs collected from a turbine system. Each record in the dataset corresponds to a fault detection event recorded by the monitoring system at a specific timestamp. It helps engineers and analysts study fault patterns, timing, and recurrence to improve maintenance scheduling, fault prediction, and system reliability.

Fault Code	Meaning (Example)	Description
GF	Generator Fault	Issues in generator components like stator or rotor.
AF	Air Flow Fault	Abnormal air intake or exhaust pressure.
TF	Temperature Fault	Overheating or temperature sensor anomalies.
VF	Vibration Fault	Excessive turbine vibration.
LF	Lubrication Fault	Lubrication oil level or pressure issue.

Code :

```
import pandas as pd

df = pd.read_csv("fault_data.csv")
df
```

	DateTime	Time	Fault
0	2014-05-14 14:39:44	1.400096e+09	GF
1	2014-05-14 14:50:24	1.400097e+09	GF
2	2014-05-14 14:58:56	1.400098e+09	GF
3	2014-05-14 15:09:36	1.400098e+09	GF
4	2014-05-14 15:20:16	1.400099e+09	GF

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 553 entries, 0 to 552
Data columns (total 3 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   DateTime  553 non-null    object
 1   Time      553 non-null    float64
 2   Fault     553 non-null    object
dtypes: float64(1), object(2)
memory usage: 13.1+ KB
```

df.describe()

	Time
count	5.530000e+02
mean	1.414452e+09
std	5.194956e+06
min	1.400096e+09
25%	1.412770e+09
50%	1.414496e+09
75%	1.418045e+09
max	1.421385e+09

df['Fault'].value_counts()

```
Fault
FF    254
EF    174
AF     62
GF     43
MF     20
Name: count, dtype: int64
```

Data Cleaning

Check for missing values, duplicates, and ensure DateTime is in datetime format.

df.isnull().sum()

```
DateTime    0
Time        0
Fault       0
dtype: int64
```

df.duplicated()

```
0      False
1      False
2      False
3      False
4      False
       ...
548    False
549    False
550    False
551    False
552    False
Length: 553, dtype: bool
```

Convert to datetime
df['DateTime'] = pd.to_datetime(df['DateTime'])

Add Time-Based Features
You can extract time-based attributes for trend analysis.

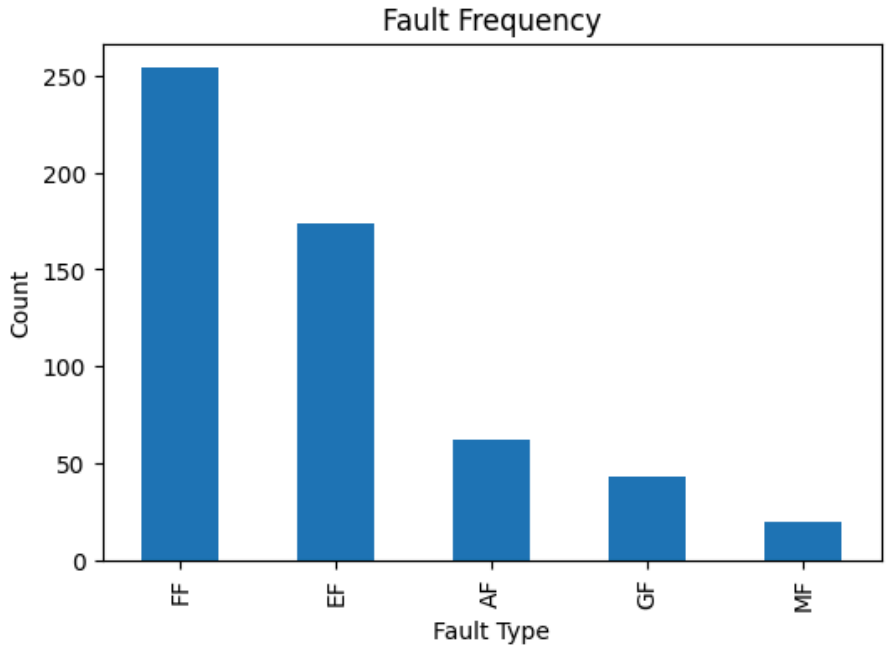
df['Year'] = df['DateTime'].dt.year
df['Month'] = df['DateTime'].dt.month
df['Day'] = df['DateTime'].dt.day
df['Hour'] = df['DateTime'].dt.hour
df['Weekday'] = df['DateTime'].dt.day_name()

	DateTime	Time	Fault	Year	Month	Day	Hour	Weekday
0	2014-05-14 14:39:44	1.400096e+09	GF	2014	5	14	14	Wednesday
1	2014-05-14 14:50:24	1.400097e+09	GF	2014	5	14	14	Wednesday
2	2014-05-14 14:58:56	1.400098e+09	GF	2014	5	14	14	Wednesday
3	2014-05-14 15:09:36	1.400098e+09	GF	2014	5	14	15	Wednesday
4	2014-05-14 15:20:16	1.400099e+09	GF	2014	5	14	15	Wednesday

Fault Frequency Analysis
See which faults occur most often.

import matplotlib.pyplot as plt

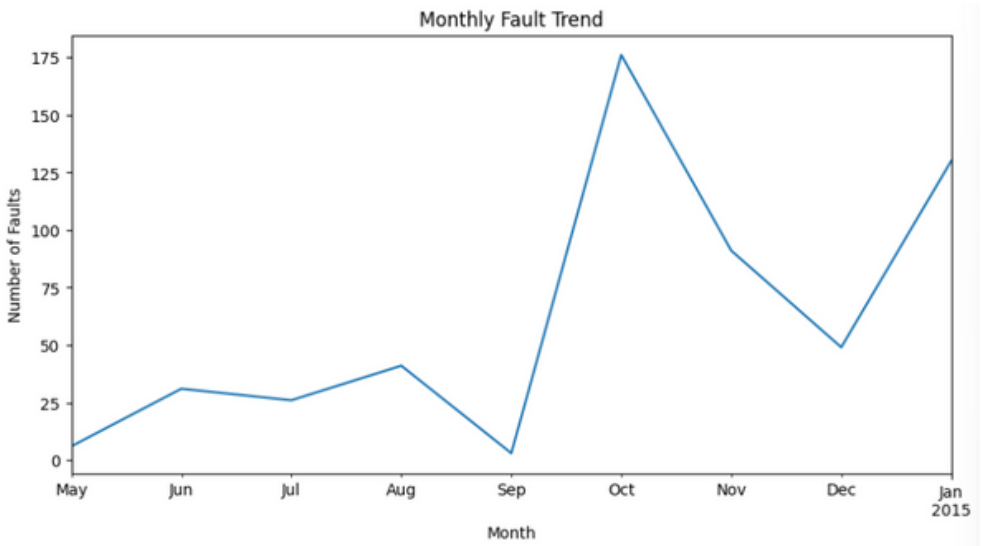
fault_counts = df['Fault'].value_counts()
fault_counts.plot(kind='bar', figsize=(6,4))
plt.title('Fault Frequency')
plt.xlabel('Fault Type')
plt.ylabel('Count')
plt.show()



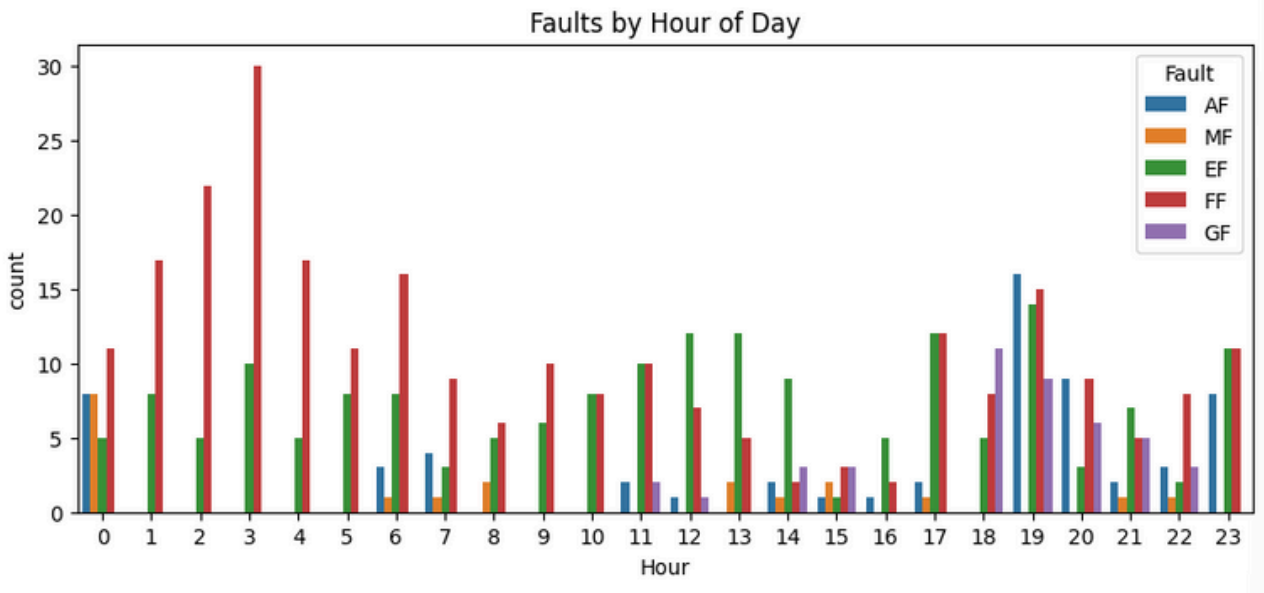
Trend Over Time
Analyze fault trends over days or months.

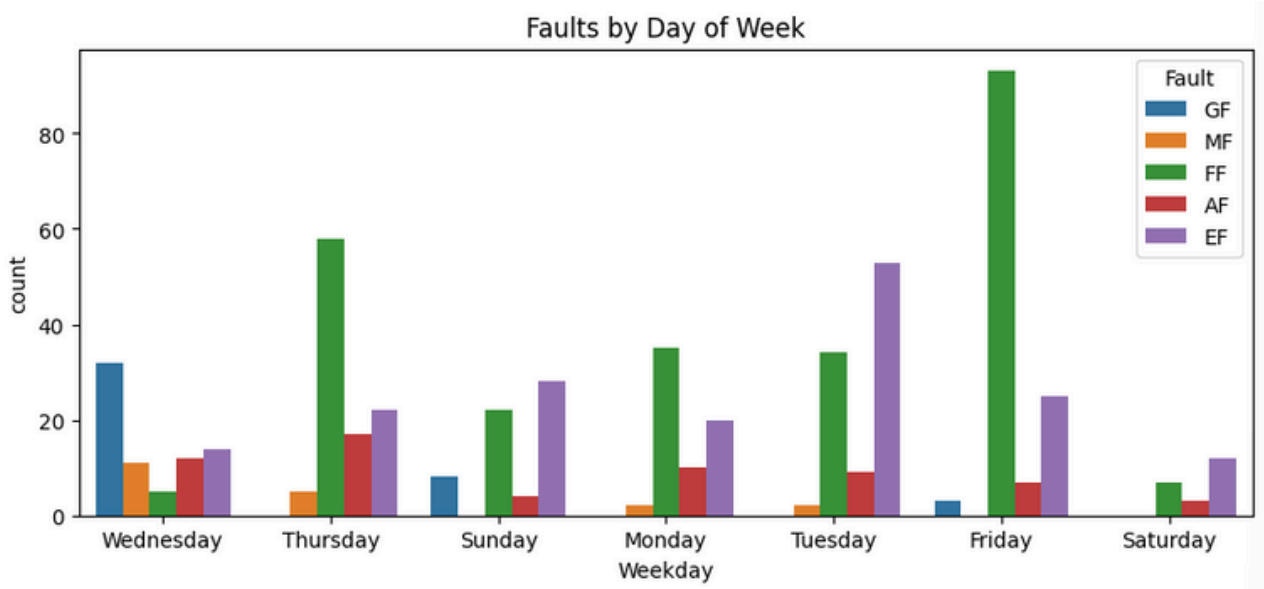
```
df.set_index('DateTime', inplace=True)
monthly_faults = df.resample('M').size()

monthly_faults.plot(figsize=(10,5))
plt.title('Monthly Fault Trend')
plt.xlabel('Month')
plt.ylabel('Number of Faults')
plt.show()
```



Fault Distribution by Hour/Weekday
Check if faults happen more during specific times.





Fault Correlation or Transition

If you have multiple fault types (GF, AF, TF, etc.), check transitions between them.

```
df['NextFault'] = df['Fault'].shift(-1)
transition = pd.crosstab(df['Fault'], df['NextFault'])
```

NextFault	AF	EF	FF	GF	MF
Fault					
AF	45	3	1	0	12
EF	4	82	86	1	1
FF	2	85	165	2	0
GF	1	2	0	39	1
MF	10	2	2	0	6